Paper outline

Testing LD clusters instead of SNPs — structure, evidence inventory, and gaps

module_sim_LDscnR

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Contents

1 The argument in one paragraph

A genome scan tests SNPs; the object of interest is a locus. Benjamini–Hochberg controls the false discovery proportion **among the units tested**, so a single-SNP scan controls it among SNPs while being judged against a locus-level truth. On one simulated panel, 2,467 significant SNPs were **132 distinct LD clusters**, with the rejected set 6.3 times more redundant than the genome as a whole — so "5% FDR over 2,467 discoveries" is a guarantee about a different object from the one being counted. Testing near-independent LD clusters aligns the two. The paper is that argument plus a pipeline that derives its units from the data's own LD decay rather than from chosen constants.

> **The central claim requires no simulation.** It follows from what BH controls. Everything measured below is supporting evidence for a claim that stands on its own, which is an unusual and exploitable position: the simulations show the size of the effect, not whether it exists.

2 Why it is one paper and not four

Each link is load-bearing rather than merely sequential. Remove any one and the argument fails at a specific point.

Link	**What fails without it**
LD-decay estimation	The clustering thresholds become chosen constants. The decay fit derives `min_r2` and the merge distance *per chromosome*, so a threshold means the same thing on chromosomes with different recombination.
ld_w	No units. ld_w is what makes a cluster a cluster — and, separately, is shown *not* to work as a filter within them, a negative that earns its place by ruling out the obvious alternative use.
Two-stage reduction	The reported FDR is over SNPs. This is the claim.
Association test	Nothing — and that is the point. EMMAX and LFMM give the same conclusions, which is what shows the result is a property of the units rather than of any test.
Consistency analysis	No guidance on which derived thresholds the answer depends on, so the pipeline is not usable by anyone else.

3 Proposed structure

3.1 1. Introduction

The redundancy problem, stated as an FDR-target mismatch rather than as a multiple-testing complaint. Existing LD-aware scans mostly prune or clump *after* testing; this reverses the order and makes the correction match the unit.

3.2 2. Methods

2.1 Decay estimation and the derived thresholds. 2.2 ld_w. 2.3 Two-stage complexity reduction. 2.4 Cluster-level testing — **Simes over members**, which beat every single-marker summary tested. 2.5 The consistency analysis.

3.3 3. Results

Section	**Headline**	**Status**
3.1 The redundancy problem	2,467 significant SNPs are 132 clusters; small-p tail 6.3× more redundant than the genome	**done**
3.2 What clustering buys	Precision 2× higher (62 of 71 panels). A median 51 of 67 single-SNP flags tag no QTN, against 7 of 10	**done**
3.3 How to summarise a cluster	Simes > eMLG > representative > min-p > highest-ld_w	**done**
3.4 The metric is a claim	F_β crossover at $\beta \approx 2$; below it clustering wins, above it does not	**done**
3.5 Engine replication	EMMAX and LFMM: same composition result, same crossover	**done**
3.6 Sensitivity	Two axes — selection (27-cell grid) and test (engine) — reported separately	**not run**
3.7 Empirical demonstration	—	**missing**

3.4 4. What does not work

Short, and it earns its place by ruling out the obvious things a reader will propose. The C-score as a ranking statistic; four filters (ld_w top-k, MAF, cluster size, geometric exclusion) all strongly enriched for QTN-bearing units and none improving precision-recall. **Enrichment predicts precision within a restriction, but no restriction beats using every cluster** — the recall lost always exceeds the precision gained.

3.5 5. Limitations

4 Parameters that transfer, and the currency each transfers in

The pipeline has three tunable thresholds. All three were being quoted as *numbers*, and measuring what each number actually does — across the eight simulation settings, and against the stickleback panel — shows that **each transfers in a different currency, and for two of them the naive choice is wrong.**

Parameter	Across sim settings	To the panel	Transfers as
`distance_threshold` 1e5	**fails**: 0.60–10.94% of flagged-cluster gaps exceed it, an 18× spread; switching to the derived value removes **68–100%** of split points in every setting	**also consequential**, in the opposite direction: derived is 3–18 kb, *below* the hand-set value	an **operating point** (gap exceedance), never a bp value — but the operating point itself does not transfer
size floor 2	**stable**: retains 46.0–51.2% of clusters across all eight settings, a 1.1× spread	**does not**: 17.0%, a 2.8× gap	a **structural value** — Simes over a singleton *is* that marker's p — with the differing retention reported as a result
`ld_w_threshold` 0.025	**fails**: retains 5.1% to 48.1% of markers, a 9.4× spread	panel's 28% sits *inside* the simulation range	**marker retention**, matched per dataset; the numeric value carries no meaning

Only the size floor transfers, and pooling hid both failures. Averaged over cells, `ld_w` retention reads 55/20/5% against the panel's 54/28/13% — close enough to call the axis well matched, which is what was reported to the panel session. Broken out, that 20% is the average of 5.1% and 48.1%.

The two failing axes are also not independent. Gaps exceeding the distance threshold track the number of flagged stage-1 clusters (Spearman −0.905), and `ld_w_threshold` is what sets that count — 1,272 to 32,318 across the eight settings. So a factorial grid cannot vary these two orthogonally and read the margins separately, which is a constraint on the sensitivity design rather than a caveat on it.

A withdrawn claim, and how it was caught. This table first read "`distance_threshold` inert everywhere, derived exceeds the largest gap on 160 of 160 chromosomes". That was computed on adjacent-marker gaps over the whole map; `split_by_distance()` uses gaps between consecutive *flagged stage-1 clusters*. Wrong unit, wrong subset, and the correct quantity reverses the conclusion. Found by the panel session reading the source rather than by any disagreement in the numbers — which is the second time today a result broke under decomposition rather than under challenge.

The practical rule that follows is **different invariants for different axes** — match marker retention on `ld_w`, hold the size floor at an identical value and report the differing retention, and specify `distance_threshold` by what fraction of gaps it splits rather than in bp. A grid whose three axes all read as matched numbers would be comparing three different things.

4.1 Why this is a result and not housekeeping

The `distance_threshold` case is the sharpest. The package derives it as $d = \rho/(a(1-\rho))$, so the window **collapses as decay speeds up** and carries no term for the size of the feature stage-2 merging exists to bridge. On the panel that yields a 6 kb median merge distance on a genome whose motivating feature is a 430 kb inversion. Here it yields roughly 1 Mb — which is *not* past the top of the flagged-cluster gap distribution, and adopting it would remove 68–100% of split points depending on the setting. Identical rule, consequential on both datasets, and in opposite directions: too small to span the features there, and physically meaningless in its tail here (4,961 Mb derived on a 30 Mb chromosome). Three fixes were proposed and all three refuted, each acting hardest exactly where the local estimate is most informative.

5 What still needs to be done

Ordered by what blocks what.

1	**Check whether the cells are comparable at all**	`ld_w_threshold` 0.025 retains 5–6% of markers in `V0.5_c1` and 26–48% in `V0.5_c2`. The cell where clustering wins 19/19 is at the most aggressive end. This does not invalidate anything — unflagged markers are still clustered and tested — but the cross-cell comparison carrying the headline result has not been checked at matched operating points. **Newly surfaced; do this before the sensitivity grid.**
2	**The factorial sensitivity grid**	$3 \times 3 \times 3$, 9 re-clusterings. Now better specified than when it was planned: report *operating points* rather than values, never pool across cells, and report the two axes (selection vs test) separately.
3	T_2 / T_2b for the local threshold	Scoped down: since the local rule collapses to a flat threshold on the panel, the most T_2 can support is *helps in the cold-block regime, inert to degenerate outside it.* Worth deciding whether that claim is wanted before spending the time.
4	**An empirical demonstration**	Unchanged and still the largest gap. See below.
5	**Decide what to do about the hardcoded 1e5**	It appears in roughly 20 scripts. It is *not* inert – that claim was withdrawn – so this is a live choice rather than documentation: adopting the derived rule would remove 68–100% of split points and repartition every result. Nothing needs re-running to keep the current results valid, but the choice has to be argued rather than inherited.

5.1 Settled since the first draft of this outline

- **`n_win_decay` = 20**, by two independent criteria — an internal consistency measure and an external one scored against the simulation's true recombination rate. Not 50, which buys nothing for 2.5 times the windows.
- **The recombination proxy is real**: Spearman +0.757 against exact truth, 20 of 20 chromosomes. The operational figure is **0.782** at $w = 20$, not the 0.830 headline, which was a composite of cold-block *detection* and continuous *tracking*.
- **The cap is generous or nothing**: the cost is monotone, so there is no optimum; 10 Mb costs 0.004 and bounds a quantity that otherwise reaches 4,961 Mb.
- **The size floor is inert above 2** ($p = 0.11$, 0.40, 0.053 for floor 2 to 8), and floor 1 to 2 *gains* recall while *losing* precision — so floor 2 keeps its structural justification but not the performance gloss.

- **The panel can falsify the proxy but cannot calibrate it**, because calibration needs variation in the cold-block regime it lacks. A property of the stickleback map, not of empirical data in general.

6 Evidence inventory: what would a referee ask for?

Gap 1 — no empirical demonstration. Everything is simulated. The natural companion is the 3sp panel, but its support statistic currently measures the *filter* rather than the test (units the filter selected but the test did not reject overlap peaks at 69.2% against the discoveries' 78.2%, $p = 0.18$), so it cannot carry a validation claim as it stands. **This is the largest gap.**

Gap 2 — the advantage is concentrated in one simulation cell. Simes beats the single-SNP scan in 19 of 19 panels in `V0.5_c1` and the paired tests in the other three cells are **null, not adverse** (26 W / 22 L, all $p > 0.4$) — several panels are ties in which neither analysis recovers anything. Must be stated in the Results, not a supplement: *decisive where there is signal, indistinguishable where there is not.*

Gap 3 — the sensitivity analysis is planned, not run, and it now has a prerequisite it did not have: the four cells sit at very different `ld_w` operating points (item 1, §5), so a grid run over them would inherit that. Design is in `TODO_sensitivity.md`, amended to report operating points rather than values and never to pool across cells.

Gap 4 — one simulation generation. Re-clustering at the newer decay settings changes nothing ($p = 0.25$ and 0.85), so this is closed for the decay parameter — but all results come from one simulator under one demographic model.

7 Claims the paper should not make

- That clustering finds more loci. It finds **fewer** and is right more often.
- That ld_w filtering helps. Four variants, none did.
- That any performance number transfers. The **ratios** are the transferable part; absolute precision is a property of this signal-to-noise regime.
- That a threshold is recommended for the consistency score. Every threshold tried on a C-like quantity failed, including the one that looked principled.
- That enrichment for containing a QTN is irrelevant to detection. It predicts precision; it just cannot be exploited by restriction.
- That a threshold means the same thing on two datasets because it is the same number. Measured, one of the three transfers, one does not, and one fails even across simulation settings.
- That raising the minimum cluster size improves precision. Measured, floor 1 to 2 *lowers* it.

8 Two methodological findings that may be worth their own space

Name the axis a stability measure is on. A consistency score computed over parameters that move only the selection measures how stable the *selection* is. "Called under every setting" reads as a claim about the finding. The second association engine is the missing test-side axis, which makes the engine sweep and the sensitivity analysis one design rather than two.

Replicate specifically where nobody doubts the result. The LFMM replication on the panel side surfaced two sampler defects that had been sitting in the original arm for a day with a manuscript being written around them. A replication that *agrees* leaves such defects in place; the disagreement is the diagnostic.

9 Provenance

Result document: `ld_clustering_result.pdf`. Full record including everything withdrawn: `development_history.pdf`. Both build from committed CSVs. Branch `null-rework-two-basis`.